

Assignment 3: Sketch image classification competition

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Abstract

This study explores efficient sketch image classification on a 500-class subset of ImageNet-Sketch by leveraging pretrained models like CLIP and EVA-CLIP alongside lightweight neural networks and scalable contrastive learning techniques.

1. Introduction

Image classification is a key task in computer vision. This work details the methodologies used for classifying a subset of 500 classes from the ImageNet Sketch dataset.

2. Methodes

The primary consideration in this study is to avoid training an architect from scratch, given the complexity associated with converging on a model utilizing 20,000 images for 500 classes.

2.1. Zero-shot learning using CLIP

My initial approach involved applying zero-shot learning. I aimed to leverage the capabilities of CLIP [2], a pre trained model, and utilized a mapping of code classes to class names that I found. This mapping was used to compute the inference step within CLIP.

2.2. EVA-CLIP

EVA-CLIP is a state-of-the-art foundation model that integrates vision and language through contrastive pretraining. In this study, I use the pretrained vision module, fine-tuned on ImageNet, as a feature extractor for image classification, focusing on training lightweight classification modules to reduce computational overhead. I chose this Model because it represents the state of the art in image classification on the ImageNet Sketch, as indicated by a benchmark found on [Hugging Face](#).

2.2.1. Lightweight Neural Network

The SketchClassifier is a two-layer neural network with a hidden layer, ReLU activation, dropout for

regularization, and a final layer for class logits, designed for classification.

2.2.2. Scalable Contrastive Learning

This method introduces a scalable Large-Scale Contrastive Learning Framework, combining an encoder with a contrastive learning head and a progressive classifier for improved discrimination. It uses Multi-Class SupCon Loss with temperature scaling to stabilize training.

2.2.3. LoRa

The method uses Low-Rank Adaptation (LoRA)[1] to efficiently fine-tune a pre-trained model by adding low-rank adapters to specific layers, such as attention layers in our case, while keeping the rest of the model frozen.

3. Experiments

3.1. Dataset

ImageNet-Sketch[3] is a domain-specific subset of ImageNet with sketch-style images across 1,000 classes, designed to evaluate model robustness to domain shifts. This challenge focuses on a 500-class subset for classification evaluation.

4. Results & Conclusion

In the initial phase of my experiments, I applied zero-shot learning using the CLIP model with a simple mapping of class names. However, the performance was suboptimal, achieving only 46% accuracy on the validation set. To improve this, I refined the prompt to: "This is a black and white sketch of a {class name}", which led to a modest increase in accuracy to 48%.

Looking ahead, we could explore the idea of prompt learning as discussed in the COOP [4](Contrastive Pretraining with Optimized Prompts) paper, which optimizes the prompt itself to better align with the task at hand. This could potentially enhance the model's performance, especially in zero-shot settings, by learning more effective and task-specific prompts.

Following the prompt refinement, I used two approaches to enhance performance: the lightweight classifier head and LoRA (Low-Rank Adaptation). With these, I achieved a significant improvement,

reaching an accuracy of 92% on the validation set. However, LoRA did not provide a

language models. In *International Conference on Learning Representations*, 2022. [1](#)

- [2] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen

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substantial improvement across all classes, and some categories, such as different dog breeds, remained difficult to distinguish, indicating challenges in classification for certain hard-to-detect classes.

To address this, I introduced a contrastive learning approach with a progressive classifier. Although I did not implement a function for active sampling of hard key queries, I relied on the random shuffling of the data loader. This process, after multiple runs, resulted in an accuracy of 94.41% on validation set (after a lot of less scores). However, I believe that leveraging an active sampling strategy for difficult queries would further improve model performance in future experiments, providing better focus on the most challenging instances.

A. Appendix: Feature Class Featureclass Similarity

This appendix includes some plots where I have visualized the similarity matrix over the validation set. We can see that there is indeed a lot of class similarity. This can help us to construct our dataloader for training more effectively, instead of relying on random chances, as I did previously.

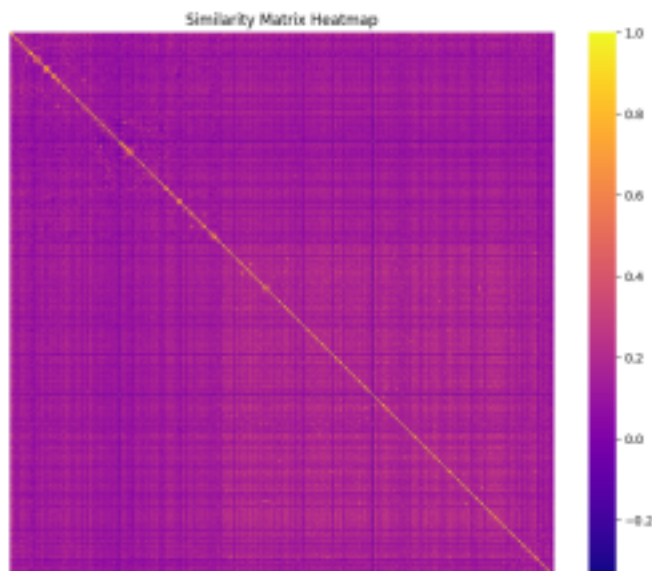


Figure 1. Similarity Matrix over the Validation Set

References

- [1] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large

Krueger, and Ilya Sutskever. Learning Transferable Visual Models From Natural Language Supervision. *arXiv e-prints*, art. arXiv:2103.00020, 2021. [1](#)

- [3] Haohan Wang, Songwei Ge, Zachary Lipton, and Eric P Xing. Learning robust global representations by penalizing local predictive power. In *Advances in Neural Information Processing Systems*, pages 10506–10518, 2019. [1](#)
- [4] Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Conditional prompt learning for vision-language models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. [1](#)